

Thirteenth International Olympiad in Linguistics

Blagoevgrad (Bulgaria), 20–24 July 2015

Individual Contest Solutions

Problem 1. Nahuatl:

- 1: *cē*, 2: *ōme*, 3: *ēyi*, 4: *nāhui*;
- 5: *mācuilli*, 10: *mahtlactli*, 15: *cactōlli*;

$$\bullet \alpha \times 20^\beta, 1 \leq \alpha \leq 5, 1 \leq \beta \leq 3:$$

α 1: <i>ceM</i> 2: <i>ōm</i> 3: <i>yē</i> 4: <i>nāuh</i> 5: <i>mācuil</i>	-	20^β 20: <i>pōhualli</i> 400: <i>tzontli</i> 8000: <i>xiquipilli</i>	;
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- 7: *chicōme*;
- $\gamma + \delta, \left\{ \begin{array}{l} \gamma \in \{10, 15\}, 1 \leq \delta \leq 4 \\ \gamma = \alpha \times 20^\beta, 1 \leq \delta < 20^\beta \end{array} \right\}: \boxed{\gamma} \text{-} oM \text{-} \boxed{\delta},$
 $M = \begin{cases} m & \text{before } m, p, \text{ or a vowel;} \\ n & \text{otherwise.} \end{cases}$

Arammba:

- 1: *ngám̐bi*, 2: *yànp̐aro*, 3: *yenówe*, 4: *asàr*, 5: *tambaroy*, 6: *nimbo*;
- $\alpha \times 6, 2 \leq \alpha \leq 5: \boxed{\alpha} \text{ tàxwo};$
- $6^2 = 36: \text{fete}, 6^3 = 216: \text{tarumba}, 6^4 = 1296: \text{ndamno}, 6^5 = 7776: \text{weremeke};$
- $\alpha \times 6^\beta, 2 \leq \beta: \boxed{\alpha} \boxed{6^\beta};$
- $\alpha \times 6^\beta + \delta, 0 < \delta < 6^\beta: \boxed{\alpha \times 6^\beta} \boxed{\delta}.$

	$\begin{array}{c} 10+1 \\ 11 \times 10 = 110 \end{array}$	(1)		$1+1 = 1 \times 2$	(7)
	$\begin{array}{c} 1 \times 20 \\ 20 \times 2 = 40 \end{array}$	(2)		$1+4 = 5$	(8)
(a)	$\begin{array}{c} 3 \times 20 + (5+2) \\ 67 + 14 = 81 \end{array}$	(3)		$\begin{array}{c} 2 \times 6 \\ 12 + 60 = 72 \end{array}$	(9)
	$5+2 = 7$	(4)		$\begin{array}{c} 3 \times 6 \\ 3 \times 18 = 54 \end{array}$	(10)
	$\begin{array}{c} 10+3 \\ 13 \times 3 = 39 \end{array}$	(5)		$6 \times 36 = 216$	(11)
	$5 \times 3 = 15$	(6)		$\begin{array}{c} 2 \times 6 \\ 6 + 12 = 18 \end{array}$	(12)

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$$\begin{array}{rcl} 3 \times 400 + 4 \times 20 + (15 + 1) & & \\ 1296 & = & 1296 \end{array} \quad (13)$$

$$\begin{array}{rcl} 1 \times 400 + 1 \times 20 + (10 + 2) & & 2 \times 216 \\ 432 & = & 432 \end{array} \quad (14)$$

$$\begin{array}{rcl} 1 \times 400 & & 216 + 5 \times 36 + 4 \\ 400 & = & 400 \end{array} \quad (15)$$

$$\begin{array}{rcl} 1 \times 8000 & & 7776 + 216 + 6 + 2 \\ 8000 & = & 8000 \end{array} \quad (16)$$

- (b)
- $42 = 2 \times 20 + 2$: *öm-pöhualli-om-öme*;
 - $494 = 1 \times 400 + 4 \times 20 + 10 + 4$: *cen-tzontli-on-nāuh-pöhualli-om-mahtlactli-on-nāhui*.
- (c)
- $43 = 36 + 6 + 1$: *fete nimbo ngámbi*;
 - $569 = 2 \times 216 + 3 \times 36 + 4 \times 6 + 5$: *yànparo tarumba yenówe fete asàr tàxwo tambaroy*.